

LicenseKaki Master Notification Catalog

Date: 2026-04-24

Position: Single Source of Truth — consolidates all notifications (role × event × channel) into one reference

Source Documents:

- Requirements & AC: [notification-requirements.md](#) (Korean)
- Strategy & Channels & Journeys: [notification-strategy.md](#) (Korean)
- Korean catalog: [notification-catalog.md](#)

Purpose: Define scope of the infrastructure sprint (SMS/WhatsApp gateway, phoneNumber JIT collection, Preference Center, Digest engine) and provide the build order reference.

0. Legend

Abbrev	Meaning
E	Email
I	In-app notification (logged-in users / Notification table)
S	SMS
W	WhatsApp Business API
★	Critical — legal/financial/security mandatory, no opt-out, exempt from Quiet Hours
•	Important — journey progress info, opt-out by category allowed
○	Informational — reassurance, digest, reference
M	Marketing — explicit opt-in only, Spam Control Act §ADV labeling required
✓	Currently implemented
Δ	Partial (some channels missing)
x	Not implemented

Common Principles (apply to every notification)

1. One message = one primary CTA (Stripe pattern)
2. No sensitive data (name, address, license #, amount) in Email subject / SMS body — route via platform link
3. Strict separation of Transactional vs Marketing — no promotional content in transactional alerts
4. In-app (I) is always created on every state transition — no opt-out (audit trail)
5. Quiet Hours (SGT 22:00–08:00): only Critical sends immediately; others queued for 08:00
6. Daily caps per user: E 5 / S 2 / W 4 — auto-switch to digest if exceeded
7. Idempotency: no duplicates within 30 min for the same `user_seq + reference_type + reference_id + type + channel`
8. Failure isolation: notification delivery failure must NEVER rollback business transaction (`afterCommit` hook + WARN log)

1. Channel Decision Matrix

Condition	Required channels	Optional
All state transitions (log)	I	—
External proof needed (payment, license, LOA)	E	—
<24h deadline + legal/financial consequence	E + I	S or W
On-site visit (Concierge, Expired License, LEW Service)	E + I	W (D-1) + S (30 min before arrival)
Signature request (LOA, e-doc)	E + I	W (reminder)
Admin internal operations	I	E (on SLA breach)
Security (password change, new device)	E + I	—
Marketing	E (opt-in)	W (opt-in)

2. APPLICANT — 54 Notifications

2.1 Account & Security

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-01	Email verification after signup	Right after signup	Self	E	★	Action	24h valid link + 1-line service intro	/verify-email?token=	✓
A-02	Welcome / onboarding	After verification	Self	E+I	○	Re-engage	"Start your first application" CTA	/ applications/new	✗ P2
A-03	Password reset link	forgot-password request	Self	E	★	Action	1h valid reset link + identity check note	/reset-password?token=	✓
A-04	Password change successful	resetPassword completed	Self	E	★	Security	"Your password was just changed. If this wasn't you..." + support contact	(none)	✗ P0
A-05	New device / IP login detected	Login + novel UA fingerprint	Self	E	●	Security	Time/IP/UA + "Not you?" link	/security/sessions	✗ P2

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-06	Inactive account activation link	Login attempt on inactive account	Self	E	★	Action	Activation link + PDPA re-consent prompt	/activate?token=	✓
A-07	Resend verification email	User request	Self	E	•	Action	New verification link	/verify-email?token=	✓

2.2 Application Main Flow

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA
A-08	Application submitted – receipt	createApplication success	Self	E+I	•	Info	Application #, address, expected LEW review window (24–72h) + PDF attached	/ap/{i
A-09	Draft saved (abandonment reminder)	D+1 / D+3	Self	E	○	Re-engage	"You have a draft in progress" – max 2 reminders	/ap/{i
A-10	LEW assigned	assignLew success	Self	E+I	•	Info	"Your reviewer {name} has been assigned" + expected timeline	/ap/{i
A-11	kVA confirmed	confirmKva	Self	E+I	•	Info	Confirmed kVA + fixed fee quote	/ap/{i
A-12	Document request created by LEW	LEW creates request	Self	E+I	•	Action	Required documents list + submission deadline	/ap/{i#o
A-13	Documents approved by LEW	LEW approval	Self	E+I	•	Info	Approved docs + remaining required list	/ap/{i#o
A-14	Documents rejected by LEW	LEW rejection	Self	E+I+S	★	Urgent action	Rejection reason + re-upload	/ap

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CT
							request + 24h SLA	{i #c
A-15	Revision requested (REVISION_REQUESTED)	requestRevision	Self	E+I (+ W opt-in)	★	Action	Full revision notes + submit CTA	/ ap {i
A-16	Revision not submitted reminder	D+2 / D+5	Self	E (+ W D+5 only)	•	Nudge	"Auto-cancelled on D+7" — max 2	/ ap {i
A-17	Payment required (PENDING_PAYMENT)	approveForPayment	Self	E+I	★	Action	PayNow UEN / reference code / amount / deadline + anti-phishing note	/ ap {i
A-18	Payment due D-3 reminder	Scheduler	Self	E	•	Nudge	Deadline restated + PayNow reminder	/ ap {i
A-19	Payment due D-1 reminder	Scheduler	Self	E+S (+ W opt-in)	★	Urgent nudge	24h countdown — SMS 160 chars, short URL	l {c
A-20	Payment confirmed (PAID)	confirmPayment	Self	E+I	•	Info	Receipt PDF + "Work will now begin"	/ ap {i
A-21	In-progress reassurance	IN_PROGRESS stable D+3	Self	E	○	Reassurance	"Your LEW is working on-site" — once, flag required	/ ap {i
A-22	License issued (COMPLETED)	completeApplication	Self	E+I+S (+ W opt-in)	★	Info critical	License #, expiry date, PDF — SMS conveys issuance fact only	/ ap {i
A-23	Admin manual status change	updateStatus	Self	I (+ optional E)	•	Info	Before/ after state	/ ap {i

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA
							+ reason (if any)	

2.3 License Expiry Lifecycle

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-24	Expiry D-90 advance notice	Scheduler	Self	E	○	Info	Expiry date + renewal guidance	/applications/{id}/renew	x P1
A-25	Expiry D-60 reminder	Scheduler	Self	E	●	Nudge	"You can renew now"	/applications/{id}/renew	x P1
A-26	Expiry D-30 warning	Scheduler	Self	E+I	●	Warning	Deadline + Expired License visit service recommendation	/applications/{id}/renew	Δ (no I)
A-27	Expiry D-7 warning	Scheduler	Self	E+W (opt-in)	★	Urgent	Expires in 1 week	/applications/{id}/renew	x P0
A-28	Expiry D-1 final warning	Scheduler	Self	E+S	★	Urgent	Expires tomorrow	lk.sg/r/{code}	x P0
A-29	License auto-EXPIRED transition	Scheduler flips to EXPIRED	Self	E+I	★	Warning	Now expired + visit renewal service CTA	/orders/expired-licence/new	x P0
A-30	Post-expiry D+1 service pitch	Scheduler	Self	E	○ (Marketing-ish)	Re-engage	Expired License Order promo — once	/orders/expired-licence/new	x P2

2.4 Kaki Concierge (v1.5)

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA
A-31	Concierge request received	notifySubmitted	Self	E+I (+ account link)	●	Info	Request code + promised 24h contact + account setup link (C3)	/concierge/requests/{code}
A-32		Manager assigned	Self	E+I	●	Info	Manager name + expected	/concierge/requests/{code}

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA
	Concierge manager assigned						contact time	
A-33	Concierge quote sent	notifyQuoteSent (after call)	Self	E+I	★	Action	Verification phrase + quote amount + PayNow	(PayNow)
A-34	Concierge LOA signature request	generateLoa	Self	E+I+S (+ W opt-in)	★	Action	72h valid signature link — SMS shortened URL only	/loa/{token}
A-35	LOA reminder	48h unsigned	Self	E+S	★	Urgent nudge	72h expiry warning	/loa/{token}
A-36	LOA proxy-upload confirmation	Manager uploads	Self	E	★	Legal notice	Upload fact + 7-day objection window	mailto:support@.
A-37	Concierge licence-fee payment request	viaConcierge + PENDING_PAYMENT	Self	E+I	★	Action	Manager name included + separate from A-17 body	/applications/{id}/payment
A-38	Concierge visit scheduled	scheduleVisit	Self	E+I+S	★	Info critical	Date/ address/ manager info + iCal (.ics) attached	/concierge/requests/{code}
A-39	Visit D-1 reminder	Scheduler 09:00 SGT	Self	S (+ W)	•	Nudge	Tomorrow's visit + manager tel: link	lk.sg/v/{code}
A-40	Visit arrival 30 min away	Manager triggers	Self	S+W	•	Real-time	"Your manager is on the way" + contact	(tel:)
A-41	Concierge visit completed (photos)	uploadVisitPhotos	Self	E+I	•	Action confirm	On-site photos + work summary + completion confirmation request	/concierge/requests/{code}/confirm
A-42		COMPLETED	Self	E+I (+ W)	•	Info		/concierge/requests/{code}

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA
	Concierge final completion						Thank-you + NPS preview	
A-43	Concierge cancelled	Manager/Admin cancels	Self	E+I	•	Info	Cancel reason + refund procedure	/concierge/requests/{code}

2.5 SLD Order

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-44	SLD quote proposed	proposeQuote	Self	E+I	★	Action	Amount / validity / PayNow	/orders/sld/{id}	x P1
A-45	SLD quote reminder	D-3 / D-1	Self	E (+ W D-1)	•	Nudge	Validity expiry warning	/orders/sld/{id}	x P1
A-46	SLD drawing uploaded	uploadSld	Self	E+I (+ W opt-in)	•	Action confirm	Preview link + review request	/orders/sld/{id}	x P1
A-47	SLD order completed	markComplete	Self	E+I	•	Info	DXF/PDF attached + usage guide	/orders/sld/{id}	x P2

2.6 Expired License Order (On-site visit)

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-48	Expired License quote proposed	proposeQuote	Self	E+I	★	Action	Quote + available visit slots	/orders/expired-licence/{id}	x P1
A-49	Expired License visit scheduled	scheduleVisit	Self	E+I+S	★	Info critical	iCal attached + manager info	/orders/expired-licence/{id}	x P0
A-50	Expired License visit D-1 reminder	Scheduler	Self	S (+ W)	•	Nudge	Tomorrow's visit confirmation	(short URL)	x P0
A-51		checkIn (manager)	Self	I	○	Info			x P2

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
	Visit check-in						"Manager has arrived"	/orders/expired-licence/{id}	
A-52	Visit completed	uploadVisitPhotos	Self	E+I	•	Action confirm	Photos / diagnosis / completion confirmation	/orders/expired-licence/{id}/confirm	x P1
A-53	Expired License final completion	Confirmation	Self	E+I	•	Info	Thank-you + receipt	/orders/expired-licence/{id}	x P2

2.7 Feedback

#	Event	Timing	Recipient	Channel	Severity	Purpose	Summary	CTA	Status
A-54	NPS survey request	Completion D+3	Self	E+I	○	Re-engage	5-point 1-click — once only	/feedback/{token}	x P2

3. LEW (Licensed Electrical Worker) — 12 Notifications

Experience principle: Workflow efficiency focus. Important/Informational → **digest at 09:00 / 15:00 SGT**, Critical only real-time.

#	Event	Timing	Channel	Severity	Digest	Purpose	Summary	Status
L-01	LEW registration approved	approveLew	E+I	★	No	Info	Dashboard CTA + onboarding guide	x P0
L-02	LEW registration rejected	rejectLew (reason required)	E	★	No	Action	Rejection reason + how to reapply	x P0
L-03	Application assigned	I immediate / E digest with 10 min debounce	I + E(Digest)	•	Yes	Work item	Application # / address / est. time	Δ (no I) P1
L-04	Unassigned	unassignLew	E+I	•	No	Info	Unassign reason	x P1
L-05	Documents uploaded	15 min debounce	I + E(Digest)	•	Yes	Work trigger	Uploaded docs list	✓

#	Event	Timing	Channel	Severity	Digest	Purpose	Summary	Status
	(review needed)							
L-06	Payment confirmed (work can start)	confirmPayment	E+I	•	No	Work trigger	Site address + contact + ready state	✓
L-07	Revision resubmission received	resubmit	I + E(Digest)	•	Yes	Work resume	Resubmitted content	x P1
L-08	SLA warning (24h no action)	Scheduler	E+I	★	No	Nudge	Pending item + deadline	x P1
L-09	SLA breach (48h+)	Scheduler	E+I (+ Admin CC)	★	No	Escalation	Notify Admin simultaneously	x P0
L-10	LEW Service Order events	Per state transition	E+I	•	Yes (non-urgent)	Field work	Booking / changes / completion	x P1
L-11	Field work scheduled (LEW → Applicant)	LEW enters date	(trigger for applicant send)	•	No	Cross-notif	—	x P1
L-12	Daily closing summary	Daily 18:00 SGT	E	○	(self)	Recap	N done today / M pending	x P2

4. ADMIN — 10 Notifications

Experience principle: Signal-to-noise is paramount. Normal flow → in-app log only. **SLA / errors / breach** → email push.

#	Event	Timing	Channel	Severity	Purpose	Summary	Status
M-01	New Application received	createApplication	I (+ optional E)	•	Work queue	App # / kVA / assignment candidate	x P0
M-02	New LEW registration	Signup with LEW role	E+I	•	Approval work	Candidate info + credential summary	x P0
M-03	New concierge request	notifySubmitted	E+I	•	Work queue	PDPA note — subject excludes	✓

#	Event	Timing	Channel	Severity	Purpose	Summary	Status
						customer name	
M-04	Concierge 24h SLA breach	Hourly scheduler	E+I	★	Escalation	Request code + Manager + elapsed hrs	✗ P0 (enum unused)
M-05	LEW SLA breach CC	Synced with L-09	E+I	★	Escalation	Application + LEW info	✗ P0
M-06	Payment / PayNow match failure	Matching scheduler	E+I	★	Anomaly	Applicant / amount / reference code	✗ P1
M-07	Invoice auto-generation failure	invoiceGenerationService	I (+ E)	•	Anomaly	Reason + retry outcome	Δ
M-08	Data breach alert	DataBreachService	E+I	★	Legal response	PDPA §26D 3-day notification obligation	✗ P1
M-09	LEW license auto-expiry detected	Scheduler	E+I	•	Admin	Expired LEW list	✗ P1
M-10	Daily operations digest	Daily 09:00 SGT	E	○	Monitoring	Intake / completion / SLA metrics	✗ P2

5. SYSTEM_ADMIN — 5 Notifications

#	Event	Timing	Channel	Severity	Purpose	Summary	Status
S-01	System failure (SMTP fail rate >5%)	Metrics threshold	E (+ optional Slack)	★	Ops alert	Recent N min fail rate + log link	✗ P1
S-02	File encryption key load failure	App startup	E	★	Ops alert	Env name + failure cause	✗ P1
S-03	AI Service long outage	Health-check scheduler	E	•	Ops alert	Container state + git commit	✗ P2
S-04	DB backup failure	Backup scheduler	E	★	Ops alert	Backup time + failure reason	✗ P1
S-05	ADMIN M-* carbon copy (optional)	Each M-* event	E+I	•	Auth backup	Same body	✗ P2

6. SLD_MANAGER — 7 Notifications

Experience principle: "What order do I work on now?" — queue processing. Digest-first.

#	Event	Timing	Channel	Severity	Digest	Purpose	Status
D-01	New SLD Order	<code>createOrder</code> 15 min debounce	I + E(Digest)	•	Yes	Work queue	x P0
D-02	Assigned as manager	<code>assignManager</code>	E+I	•	No	Work assignment	x P1
D-03	Payment complete (start work)	<code>acceptQuote</code> → PAID	E+I	•	No	Work trigger	x P0
D-04	Quote rejected by applicant	<code>rejectQuote</code>	E+I	•	Yes	Lost opportunity	x P1
D-05	Revision requested	<code>requestRevision</code> (SLD)	E+I	•	No	Work resume	x P0
D-06	Applicant completion confirmed	<code>confirmCompletion</code>	I	○	Yes	Record only	x P2
D-07	Daily queue summary	Daily 08:00 SGT	E	○	(self)	Queue visibility	x P2

7. CONCIERGE_MANAGER — 9 Notifications

Experience principle: Field-mode — highest WhatsApp usage. Visit-related immediate, others digest.

#	Event	Timing	Channel	Severity	Digest	Purpose	Status
C-01	New concierge request	<code>notifySubmitted</code>	E+I	•	No	Work queue	✓
C-02	Assigned as manager	Manager designated	E+I	•	No	Work assignment	x P1 (enum unused)
C-03	24h first-contact SLA imminent/ breach	Scheduler	E+I	★	No	Escalation	x P0 (enum unused)
C-04	Applicant signed LOA	<code>LoaService.sign</code>	E+I	•	No	Next-step trigger	x P1
C-05	Expired License Order received	<code>createOrder</code>	E+I	•	No	Work queue	x P0
C-06	Expired License revisit requested	<code>requestRevisit</code>	E+I	★	No	Work resume	x P0
C-07			I		Yes	Record	

#	Event	Timing	Channel	Severity	Digest	Purpose	Status
	Expired License applicant confirmed	confirmCompletion		○			x P2
C-08	Daily visit summary	Daily 08:00 SGT	E (+ optional W)	○	(self)	Field planning	x P1
C-09	Visit arrival 30min trigger	Manager button	(triggers Applicant S+W)	●	No	Real-time	x P1

8. Scheduled / Reminder Notifications

Scheduler	Period	Recipients	Linked IDs	Implementation
LicenseExpiryScheduler	Daily	Applicant	A-24 D-90, A-25 D-60, A-26 D-30, A-27 D-7, A-28 D-1, A-29 EXPIRED, A-30 D+1 pitch	Separate flags (expiryNotifiedAt90/60/30/7/1 , expiredNotifiedAt)
PaymentDueScheduler	Daily	Applicant	A-18 D-3, A-19 D-1	Based on Application.paymentDueAt
RevisionReminderScheduler	Daily	Applicant	A-16 (D+2, D+5)	Based on revisionRequestedAt , max 2
ConciergeSlaScheduler	Hourly	Admin + Concierge Manager	C-03 / M-04	First ConciergeNote presence
LewSlaScheduler	Hourly	LEW + Admin	L-08 (24h), L-09 (48h)	Based on assignedAt
VisitReminderScheduler	Daily 09:00 SGT	Applicant	A-39, A-50 (D-1 visit)	Based on ConciergeRequest/ ExpiredLicenseOrder.visitAt
LoaReminderScheduler	Hourly	Applicant	A-35 (48h unsigned)	Based on Loa.generatedAt
DigestScheduler	Daily 09:00 /	LEW, SLD_MANAGER, CONCIERGE_MANAGER	L-03, L-05, L-07, D-01,	notification_digest_batch bundling

Scheduler	Period	Recipients	Linked IDs	Implementation
	15:00 SGT		D-04, D-06, C-07	
DailySummaryScheduler	Daily 08:00 SGT	SLD_MANAGER, CONCIERGE_MANAGER	D-07, C-08	Aggregates queue / in-progress / done
LewDailySummaryScheduler	Daily 18:00 SGT	LEW	L-12	Today done / pending
AdminDailyDigestScheduler	Daily 09:00 SGT	ADMIN	M-10	KPI snapshot
SystemHealthScheduler	Every 15 min	SYSTEM_ADMIN	S-01 (SMTP), S-03 (AI), S-04 (DB backup)	Threshold-based

9. Digest Bundling Rules

Role	Bundled events	Debounce	Send time (SGT)	Cap	Email subject
LEW	L-03, L-05, L-07, L-10 (non-critical)	10–15 min	09:00 / 15:00	50 per batch	[Digest] N review tasks today
SLD_MANAGER	D-01, D-04, D-06	15 min	09:00 / 15:00	30 per batch	[Digest] SLD order queue
CONCIERGE_MANAGER	C-07 + non-field events	15 min	15:00	20 per batch	[Digest] Concierge workload
ADMIN	M-03, M-07 (non-critical)	30 min	09:00	—	[Digest] Ops summary

Critical exception: ★ events bypass debounce and send immediately.

10. Category × Channel Opt-Out Matrix (Applicant default)

Category	In-app	Email	SMS	WhatsApp	Example IDs
SECURITY	 ON	 ON	optional	—	A-04, A-05
STATUS	 ON	 ON	—	optional	A-08, A-10, A-23
PAYMENT	 ON	 ON	optional	optional	A-17, A-19, A-20, A-37

Category	In-app	Email	SMS	WhatsApp	Example IDs
REMINDER	ON/OFF	ON/OFF	ON/OFF	ON/OFF	A-16, A-18, A-35, A-45
VISIT	 ON	 ON	optional	optional	A-38, A-39, A-40, A-49, A-50
REASSURANCE	ON/OFF	ON/OFF	—	—	A-21
EXPIRY	 ON	 ON	—	optional	A-24–A-29
MARKETING	—	OFF (default)	OFF (default)	OFF (default)	A-30 (borderline), newsletter
FEEDBACK	ON/OFF	ON/OFF	—	—	A-54

 = Critical/Transactional, no opt-out. Complies with PDPA §13 / Spam Control Act.

11. Content Template — Common Elements

All templates follow this structure:

Email

- From: LicenseKaki <noreply@licensekaki.sg> (reserved address)
- Reply-To: by category (support@ , concierge@ , billing@)
- Subject: no sensitive data — reference code only, e.g. [LicenseKaki] {event} · #{publicCode}
- Body blocks: ① greeting ② 1-sentence headline ③ detail ④ **single primary CTA button** ⑤ secondary link (dashboard) ⑥ footer (address / opt-out link / anti-phishing)
- Footer opt-out: omitted on legally-required notifications

SMS (≤160 chars)

- Format: [LicenseKaki] {action} {detail}. {shortUrl}
- Example: [LicenseKaki] Payment due 25 Apr. Pay: lk.sg/p/A1B2
- No sensitive data — short URL routes to platform
- "Reply STOP to unsubscribe" required on marketing SMS

WhatsApp (Business Template)

- Only Meta-pre-approved templates
- Rich format: image/PDF + up to 3 buttons
- Outside the 24h conversation window, only templates work
- Two-way conversation is Phase 2 (Concierge Manager only)

In-app

- Title: i18n keyed by type
- Body: ≤100 char snippet
- Metadata: referenceType , referenceId required (for deep-link)
- Read tracking (readAt)

12. Build Priority Summary

P0.5 — Infrastructure First (MUST precede P0)

- [] **SMS gateway** — Twilio or AWS SNS, `SmsService` interface mirroring `EmailService`
- [] **WhatsApp Business API** — Meta Cloud API or Twilio, `WhatsAppService` interface, pre-approve templates
- [] **User.phoneNumber column + JIT capture** — prompt during Expired License / LOA / visit flows
- [] **Preference Center data model** — `user_notification_preferences`, `notification_category`, `notification_severity`
- [] **Digest engine** — `notification_digest_batch` table + `DigestScheduler`
- [] **Quiet Hours queuing** — time-of-day check before dispatch + queuing logic
- [] **Idempotency key** — `idempotency_key` column + uniqueness index
- [] **notification_delivery log** — per-channel send/failure/delivery-report rows
- [] **iCal generator util** — for visit emails

P0 — Legal / Payment / Security / SLA Critical (right after infra)

Group	IDs
Application core	A-08 (receipt), A-17 (pay-req I), A-19 (D-1 SMS), A-20 (confirm I), A-22 (issued S+I)
Security	A-04 (pw-change notice)
LEW mgmt	L-01, L-02, L-09 (+ M-05)
Admin queue	M-01, M-02, M-04
Expiry	A-27 (D-7), A-28 (D-1), A-29 (EXPIRED transition)
Concierge urgent	A-34 (LOA), A-35 (LOA reminder), A-37 (pay-req), A-38 (visit sched), A-39 (D-1 reminder)
Expired License	A-49 (visit sched), A-50 (D-1), C-05, C-06
SLD	D-01, D-03, D-05

P1 — Operational Improvements

- Application: A-10, A-11 (I boost), A-14 (S boost), A-15 (I boost), A-16, A-18, A-23, A-24–A-26
- LEW: L-03 (I), L-04, L-07, L-08, L-10, L-11
- Concierge: A-32, A-40, A-41, A-42, A-43, C-02, C-04, C-08, C-09
- SLD: A-44, A-45, A-46, D-02, D-04
- Expired License: A-48, A-52
- Admin: M-06, M-08, M-09
- SYSTEM_ADMIN: S-01, S-02, S-04

P2 — UX Extras / Options

- A-02, A-05, A-09, A-21, A-30, A-47, A-51, A-53, A-54

- L-12, D-06, D-07, C-07, M-07, M-10, S-03, S-05
- User-facing Preference Center UI

13. Totals

Role	Notification count
APPLICANT	54
LEW	12
ADMIN	10
SYSTEM_ADMIN	5
SLD_MANAGER	7
CONCIERGE_MANAGER	9
Grand total	97 (including overlaps & cross-role)

Currently implemented: 22 full (✓), 6 partial (Δ) — roughly **23%** of catalog

New needed: 69 (P0: 27 / P1: 30 / P2: 12)

14. Reference Documents

- PM spec (AC & technical requirements): [notification-requirements.md](#) (Korean)
- Strategy (channel / journey / benchmarks): [notification-strategy.md](#) (Korean)
- Korean catalog mirror: [notification-catalog.md](#)
- Concierge v1.5 PRD §6: `doc/Project Analysis/kaki-concierge-service-prd.md`
- LEW Service redesign: `doc/Project Analysis/lew-service-visit-redesign-spec.md`
- Current EmailService interface: `blue-light-backend/src/main/java/com/bluelight/backend/api/email/EmailService.java`
- Current NotificationType enum: `blue-light-backend/src/main/java/com/bluelight/backend/domain/notification/NotificationType.java`
- Reference Notifier pattern: `blue-light-backend/src/main/java/com/bluelight/backend/api/document/DocumentRequestNotifier.java`